

Co-Designing an Integrated Multimodal Architecture: Toward Data-Driven Adaptive Child–Robot Interaction

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Abstract—Social robots have demonstrated considerable potential to support education for children by promoting engagement and skill development. However, many existing systems still rely on Wizard-of-Oz operation, limiting their autonomy, scalability, and long-term deployment in real-world settings. This work presents the co-design of an integrated multimodal architecture that enables semi-autonomous interaction for a social robot in educational settings while establishing the sensing and interaction framework required for future adaptive AI. The proposed system integrates vision-based perception, tactile sensing for physical human–robot interaction, distance sensing for proxemic-aware behaviour adaptation, speech recognition based on local keyword detection, and teacher and child inputs through a tablet interface. These interaction data are fused by an interaction manager that adapts the robot’s behaviour according to the selected activity (ten co-created educational and socio-emotional games) and the sensing inputs for a future medium-term classroom deployment. Finally, this work also discusses the next steps in using the multimodal layer and integrating foundation models for data-driven adaptive interactions, as well as the practical challenges associated with their integration, including response latency, the privacy of sensitive child data, internet connectivity, and maintaining the decision-making within teacher-defined educational activities.

I. INTRODUCTION

Socially assistive robots (SARs) are increasingly deployed in healthcare, education, and assistive settings, where they engage in close and prolonged interactions with users [1]. Designing adaptable activities that respond to users’ needs has been shown to improve acceptance, engagement, and intention to use across different populations, including autistic children and educational communities [2], [3].

To achieve this adaptability, participatory design (PD) has become an effective methodology for developing technologies that reflect users’ real needs and preferences [2]. In social robotics, PD has involved autistic children and teachers in co-designing educational robots [3]. Despite these advances, many educational SAR applications continue to focus on one-to-one interventions rather than supporting inclusive activities involving neurotypical and neurodiverse children in mainstream classrooms [4].

Most existing SAR systems rely on the Wizard-of-Oz (WoZ) paradigm, in which a hidden operator controls the robot’s behaviour [5]. Although WoZ facilitates the evaluation of the interaction strategies, it limits scalability, and the robot’s behaviour may unintentionally depend on the operator’s interpretation [5]. Semi-autonomous approaches partially address these limitations by allowing autonomous execution of predefined activities while maintaining teachers’ supervision when necessary by WoZ [6], [7].

In child–robot interaction (CRI), spontaneous behaviours such as movement, speech, physical contact, and varying attention require robots to perceive multiple interaction cues simultaneously [8]. Consequently, robust multimodal sensing is essential for understanding user engagement, adapting robot behaviour, and ensuring safe CRIs in real-world environments [9]. Rather than relying on a single sensing modality, adaptive social robots increasingly require the integration of heterogeneous data streams to provide robust and context-aware decision-making [10].

In addition, deploying AI in child-centred environments raises important concerns regarding data privacy, child emotional development, and system robustness [11]. Throughout the PD process, teachers and parents consistently highlighted the need to protect children’s sensitive data and ensure safe interactions. Consequently, the current perception modules are integrated into the robot and process interaction data locally, avoiding the transmission of sensitive information while complying with the school’s data protection policies and ethical approval requirements. Nevertheless, recent advances in large foundation models that support privacy-preserving deployments offer opportunities to improve adaptive CRI [12]. Their integration should be carefully investigated to balance the benefits of data-driven decision-making with challenges such as response latency, privacy, internet connectivity, and maintaining teacher-defined educational boundaries. Parental and teachers’ concerns regarding AI and data sharing in SARs remain an important consideration that must be addressed through evidence-based deployments, particularly in CRIs in educational settings [13].

Therefore, this work presents the ongoing development of a multimodal sensing system integrated on a social robot co-designed to support inclusive activities. The proposed framework integrates AI perception modules to adapt the robot’s behaviour during ten co-created educational and socio-emotional games (Figure 1). By combining multimodal sensing with a modular interaction architecture, the system aims to provide semi-autonomous activities while preserving

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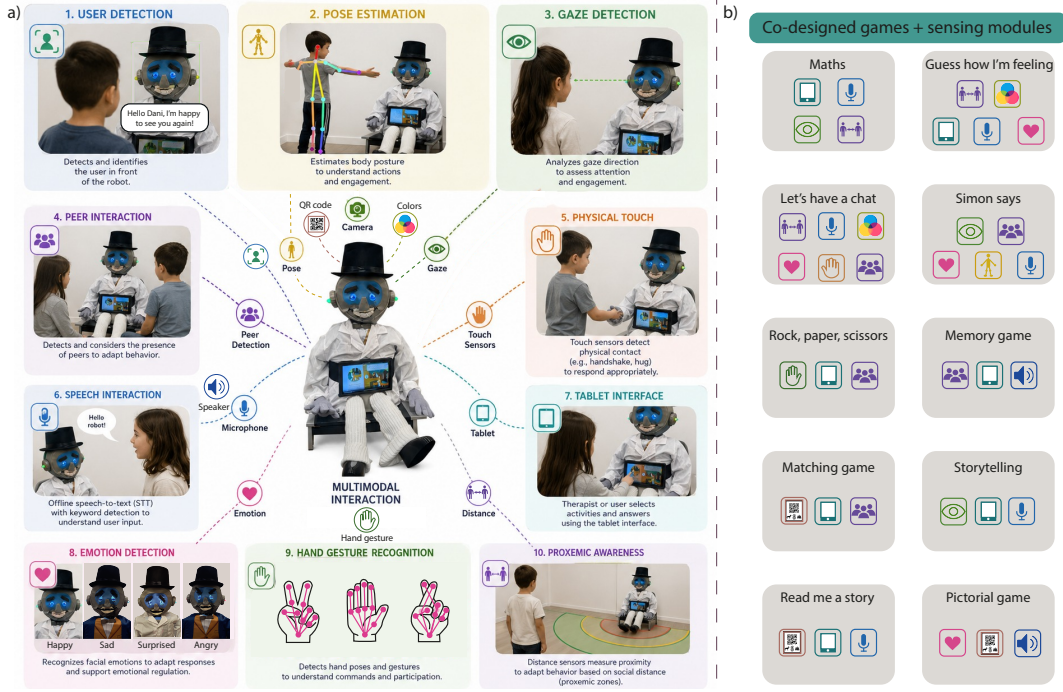


Fig. 1. Proposed framework. (a) Integrated multimodal interaction architecture, showing the AI perception modules. (b) Ten co-created activities and their associated perception modules.

teachers' supervision and supporting inclusive interactions between neurotypical and neurodiverse children in mainstream classrooms. Furthermore, the proposed architecture establishes the sensing and interaction framework required to collect multimodal data during medium-term classroom deployment, enabling the discussion for future research on data-driven adaptive CRI through reinforcement learning (RL) and foundation models (FM).

II. PARTICIPATORY DESIGN

The PD process was conducted in collaboration with a UK primary school to identify user requirements and guide the development of the robot's multimodal interaction capabilities. The process comprised three stages. First, an initial co-design session was held with teachers to identify desirable improvements and define the functionalities to be implemented in a pilot study. Second, the robot was evaluated during a single-session pilot with children, from which feedback was collected from students and teachers. Finally, following an internal brainstorming session within the research team, a second co-design session with the school was conducted to define the requirements for a medium-term deployment protocol in the classroom. The multimodal sensing functionalities presented in this work were derived from the needs identified throughout this iterative PD process.

A. Initial Co-design Session

Twelve staff members from a UK primary school participated in semi-structured interviews to identify desirable improvements to the CASTOR social robot for use with Year 2 children (6–7 years old). The session consisted of two parts: (i) a demonstration of the robot and its existing

functionalities, and (ii) semi-structured interviews exploring participants' first impressions, perceived classroom applications, usability, potential risks, and suggestions for future improvements. All participants had experience teaching children aged 4–11 years and were familiar with everyday classroom routines and learning requirements; however, none had previous experience using SARs in educational practice.

The resulting analysis included what teachers liked about the robot, what they disliked, and ideas to improve it to be implemented in a single session in their classrooms. From those ideas, the updates to the robot were developed, as well as the activities to be conducted during the pilot.

B. Second Co-Design Iteration with Children

Following the initial co-design session with teachers, a pilot study was conducted to obtain feedback from children after experiencing a real CRI in an educational setting. Although the pilot outcomes are not the focus of this work, the activity represented the second iteration of the PD process by incorporating children's perspectives into the robot's development. The pilot involved 88 Year 2 children, divided into six groups of approximately 15 participants following the same interaction protocol. Before interacting with the robot, the children completed a show-of-hands questionnaire to assess their initial perceptions regarding the robot, ease of use, social acceptance, and excitement. The interaction session comprised four phases: (i) an introductory conversation with simple engagement questions; (ii) a Simon Says activity led by the robot; (iii) a storytelling activity supported by multimedia content on the tablet, followed by comprehension questions; and (iv) a feedback and goodbye hug interaction, during which the children repeated the

show-of-hands questionnaire, provide feedback, and teachers evaluated the interaction using the UTAUT model.

Following the interaction, 27 improvement suggestions were collected from children and teachers and subsequently analysed by the research team. Twenty-three of these suggestions were translated into functional and hardware requirements that informed the next iteration of the system, including the educational activities, multimodal sensing capabilities, robot appearance, and interaction behaviours presented in Figure 1.

C. Third Co-Design Session: Medium-Term Study Design

Following the pilot study, the research team conducted a brainstorming session based on medium- and long-term deployments of social robots in educational settings, analysing intervention protocols, interaction strategies, and evaluation methods reported in the literature. Together with the feedback obtained during the pilot, these findings supported a third co-design session with six members of the partner school's staff working with Year 2 children. Based on their classroom experience and observations from the pilot, the researchers and teachers collaboratively defined the requirements for a medium-term deployment protocol that balanced the research objectives with the school's daily routines. This process guided the selection of the educational activities, the multimodal and semi-autonomous interaction requirements.

III. ACTIVITY DESIGN

The co-design generated a medium-term protocol to implement the social robot within the school's existing classroom routine. The intervention is planned around the Busy Time period, during which the robot acts as one of several optional activity stations. During the co-design process, both teachers and students highlighted the importance of appearance adaptation and increasing the robot's autonomy. Teachers expressed the need to reduce their workload while encouraging collaborative learning and socio-emotional development, whereas children requested more natural interactions, including the ability for the robot to understand spoken responses and react appropriately.

The proposed deployment involves the three Year 2 classrooms (approximately 90 students), with the robot available twice per week in each class. Interaction follows a free-choice model, allowing children to decide whether to interact with the robot individually or in groups of up to fifteen participants. This approach aims to encourage spontaneous social interaction while evaluating the robot in a realistic classroom environment. Ten activities are being developed in the system, each requiring a different combination of perception modules. Figure 1(b) illustrates the relationship between the activities and the multimodal sensing capabilities integrated into the robot. The activities are described as follows:

- **Maths:** Multiplication exercises with progressive difficulty, supported by adaptive feedback to reinforce concepts, while gaze and proxemic behaviours are monitored as engagement indicators.

- **Guess How I'm Feeling:** Emotion-recognition game where children identify the robot's emotions while facial expression analysis and emotion self-reporting assess affective state and promote emotional awareness.
- **Let's Have a Chat:** Guided conversations using speech recognition to support communication, social interaction, and encourage physical interactions such as hugs.
- **Simon Says:** Instruction-following game using pose, emotion, and speech recognition to monitor participation and peer interaction, fostering symbolic thinking.
- **Rock, Paper, Scissors:** Hand-gesture-based turn-taking activity where children's choices are recognised and the robot's selection is displayed on the tablet, fostering child-robot trust and second-order theory of mind.
- **Memory Game:** Collaborative tablet-based memory game with positive verbal reinforcement with academic-themed cards to support cognitive development.
- **Matching Game:** Activity using QR-code detection and tablet interaction to associate objects with related concepts, supporting relational categorisation skills.
- **Storytelling:** Interactive storytelling activity incorporating gaze and emotion monitoring, followed by related comprehension questions to evaluate attention span, cognitive recall, and emotional development.
- **Read Me a Story:** Assisted reading activity using keyword detection and engagement cues to provide adaptive encouragement based on reading accuracy, with communication card support for non-verbal children.
- **Pictogram Game:** Communication activity based on the Picture Exchange Communication System (PECS), enabling children to communicate through pictograms, supporting communication and sequential thinking.

IV. SYSTEM ARCHITECTURE

The multimodal interaction architecture is illustrated in Figure 1(a). The system integrates complementary sensing modalities through an embedded AI camera, microphone, touch sensors, distance sensors, and tablet interface, capturing different interaction data during CRIs. The multimodal perception layer estimates user identity, emotional state, gaze direction, body pose, hand gestures, user position, and speech and QR-code recognition. These variables are integrated by an interaction manager, which combines the user state, the selected activity, and teacher inputs to coordinate the robot's behaviour while maintaining teacher supervision and control.

The current system supports semi-autonomous interaction through predefined decision logic while continuously recording multimodal interaction data. The interaction pipeline is illustrated in Figure 2(a). Although the current architecture relies on rule-based decision making, the interaction data collected during future medium-term deployments may provide valuable insights into children's interaction patterns and could support the investigation of RL and FM approaches for adaptive CRI. Figure 2(b) illustrates one possible evolution of the current architecture, highlighting how learning-based modules could be incorporated while preserving teacher supervision and the existing multimodal perception framework.

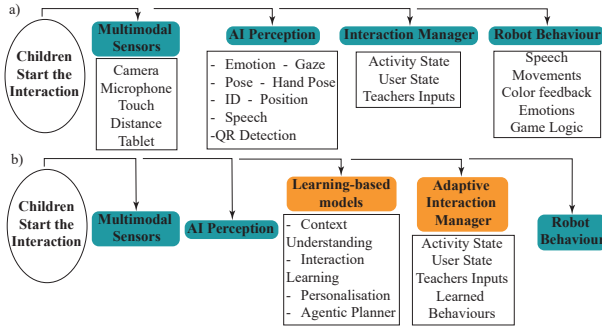


Fig. 2. System architecture and modules flux. (a) Current architecture for the medium-term deployment. (b) Potential evolution of the architecture incorporating learning-based modules.

V. CURRENT DEVELOPMENT AND DISCUSSION: TOWARDS LEARNING-BASED ADAPTIVE SOCIAL ROBOTS

The robot has been progressively upgraded by integrating and validating each sensing modality independently. Touch sensing, vision, proxemic estimation, speech recognition¹, and tablet interaction have been implemented and evaluated individually before integration into the multimodal interaction framework [14], [15], [16]. The system has the perception modules and is being combined with the ten co-created activities through a rule-based interaction manager. Each activity is being evaluated through iterative functional testing with volunteer university students before teacher usability evaluation. These tests aim to validate interaction logic and identify implementation issues before school deployment.

The current architecture relies on predefined interaction logic to ensure predictable robot behaviour and maintain teacher supervision. Although this approach facilitates trust and deployment in educational settings, it also limits the robot's ability to adapt to the wide variety of spontaneous interactions that naturally emerge during CRI. Recent advances in learning-based models and agentic behaviours suggest new opportunities for adaptive SARs. Instead of relying exclusively on predefined behaviours, these approaches can learn from multimodal behavioural data, allowing robots to personalise responses and manage situations that were not explicitly anticipated during system design.

Narayanswamy *et al.* recent work demonstrates how FMs such as Gemini can integrate heterogeneous multimodal data into unified representations that support reasoning across healthcare applications [12]. Although developed for wearable sensing, similar concepts could be explored for CRI by collecting the sensors' data of the robot and generate richer contextual representations. In this area, Suneesh is currently developing a multimodal explainability system in CRI aiming to implement an autonomous robot using multimodal cues perceived during interactions for behavioural decisions [17].

Likewise, recent studies have highlighted that deploying learning-based models in educational SARs requires careful consideration of ethical and practical constraints [11]. Teachers expressed concerns regarding data protection, predictable

behaviour, and maintain educational boundaries during the co-design sessions, as well as reported by Perella-Holfeld *et al.*, analysing parental and educators concerns about multiple areas, such as adult decision-making, socio-emotional development, privacy policies, and accessibility [13].

Within this context, the architecture presented in this work could provide an opportunity to investigate learning-based approaches in future project stages. The multimodal interaction data collected during the planned medium-term deployment may support future research on adaptive AI models in CRIs. Nevertheless, significant challenges remain, including response latency, privacy-preserving management of sensitive child data, internet connectivity in schools, and preserving teacher supervision by constraining robot behaviour within predefined educational activities.

REFERENCES

- [1] C. A. Cifuentes, M. J. Pinto, and *et al.*, "Social robots in therapy and care," *Current Robotics Reports*, vol. 1, pp. 59–74, 2020.
- [2] A. A. Ramírez-Duque *et al.*, "Collaborative and inclusive process with the autism community: A case study in colombia about social robot design," *International Journal of Social Robotics*, 2021.
- [3] N. Newbutt, L. Rice, S. Lemaignan, J. Daly, V. Charisi, and I. Conley, "Co-designing a social robot in a special educational needs school: Listening to the ambitions of autistic children and their teachers," *Interaction Studies*, vol. 23, no. 2, pp. 204–242, 2022.
- [4] B. Robins, K. Dautenhahn, R. T. Boekhorst, and A. Billard, "Robotic assistants in therapy and education of children with autism: Can a small humanoid robot help encourage social interaction skills?" *Universal access in the information society*, 2005.
- [5] E. Torubarova, J. Miniota, and A. Pereira, "Users and wizards in conversations: How woz interface choices define human-robot interactions," *arXiv preprint arXiv:2603.28338*, 2026.
- [6] A. Causo, P. Z. Win, P. S. Guo, and I.-M. Chen, "Deploying social robots as teaching aid in pre-school k2 classes: A proof-of-concept study," in *2017 IEEE International Conference on Robotics and Automation (ICRA)*, IEEE, 2017, pp. 4264–4269.
- [7] A. Pan *et al.*, "A good teacher for kids? exploring children's perceptions of mirrly, a social robot for vision health literacy education," in *Proceedings of the 25th Annual ACM Interaction Design and Children Conference*, 2026, pp. 1209–1218.
- [8] M. Escobar-Planas, V. Charisi, and E. Gomez, "that robot played with us!" children's perceptions of a robot after a child-robot group interaction," *Proceedings of Human-Computer Interaction*, 2022.
- [9] N. Efthymiou *et al.*, "Childbot: Multi-robot perception and interaction with children," *Robotics and Autonomous Systems*, 2022.
- [10] D. Zhao *et al.*, "Research status of elderly-care robots and safe human-robot interaction methods," *Frontiers in Neuroscience*, 2023.
- [11] I. R. Berson, M. J. Berson, and W. Luo, "Innovating responsibly: Ethical considerations for ai in early childhood education," *AI, Brain and Child*, vol. 1, no. 1, p. 2, 2025.
- [12] G. Narayanswamy *et al.*, "Towards a general intelligence and interface for wearable health data," *arXiv preprint arXiv:2605.22759*, 2026.
- [13] F. Perella-Holfeld, S. Sallam, J. Petrie, R. Gomez, P. Irani, and Y. Sakamoto, "Parent and educator concerns on the pedagogical use of ai-equipped social robots," *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies*, 2024.
- [14] M. Gaitán-Padilla *et al.*, "Teacher-informed tactile sensing upgrade for child-robot physical interaction: A comparison of velostat and fbg sensors," in *(Accepted) International Conference on Biomedical Robotics and Biomechatronics (BioRob)*, IEEE, 2026.
- [15] E. Sánchez *et al.*, "Proxemics-based interaction: Adapting castor's behavior through proximity sensing," in *(Accepted) International Conference on Social Robotics (ICSR)*, Springer, 2026.
- [16] M. Gaitán-Padilla *et al.*, "Enabling face tracking in an affordable social robot using a computer vision camera," in *Brazilian Symposium on Robotics (SBR)*, IEEE, 2025.
- [17] S. Suneesh, "Multimodal explainability in child-robot interaction," in *Proceedings Interaction Design and Children*, 2025.

¹<https://github.com/alphacep/vosk-api>